

POTHOLE DETECTION USING DEEP LEARNING ALGORITHMS (YOLO FAMILY)



By

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A

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ABSTRACT

Pothole detection is a very helpful technique for creating methods for the offline or real-time detection of potholes, to support real-time control of a vehicle, or to support offline data gathering for road maintenance. To distinguish between non-potholes and potholes, road pothole recognition is necessary. Identifying potholes is a difficult task. Traditional machine learning methods rely on image-based characteristics like edges and texture descriptors, such as Support Vector Machines, Principal Component Analysis, etc. Deep learning algorithms and Convolutional neural networks have shown improved performance in object recognition in recent years. Initially, this thesis work compares the different YOLO models (YOLOv3, YOLOv3-Tiny, YOLOv4, and YOLOv4-Tiny) for pothole detection and YOLOv4 gives the better result with a mean average precision of 83.44%. Secondly, this thesis work proposed the new modified YOLOv4 version to localize and detect the pothole and improved the mean average precision with 89.22% which is nearly 5% better than the original YOLOv4 model. Additionally, the modified YOLOv4 model compares to recent studies that used the YOLO variants for a similar pothole detection task and provide superior results when compared to those studies.

Keywords: *YOLOv3, YOLOv3 Tiny, YOLOv4, YOLOv4 Tiny, Machine Learning, Object Detection, Pothole Detection, Mean Average Precision*

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ANN	Artificial Neural Network
BoF	Bag of Freebies
BoS	Bag of Species
CIoU	Complete IoU
CNN	Convolutional Neural Network
CSP	Cross Stage Partial
FN	False Negative
FP	False Positive
FPN	Future Pyramid Network
GPU	Graphical Processing Unit
IoU	Intersection Over Union
mAP	Mean Average Precision
ML	Machine Learning
NA	Not Available
ReLU	Rectified Linear Unit
R-CNN	Region Convolutional Neural Network
ROI	Region of Interest
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
YOLO	You Only Look Once

CHAPTER 1: INTRODUCTION

Deep learning is a new form of machine learning (ML) and artificial intelligence (AI) that replicate the way humans brain gain the certain types of information/knowledge. In a data science field, deep learning is most important part which helps to calculate statistics and predictive modeling. Deep learning algorithms are very useful for collecting and analyzing the large amount of data with process faster and easier. The fundamental benefit of deep learning over machine learning (ML) is that the so-called feature extraction is unnecessary.

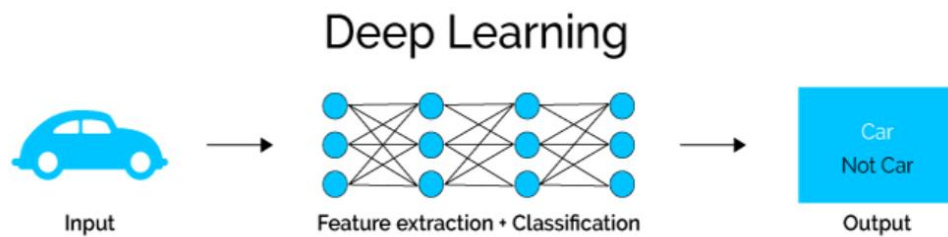


Figure 1.1 Feature extraction in deep learning (Oppermann, 2020)

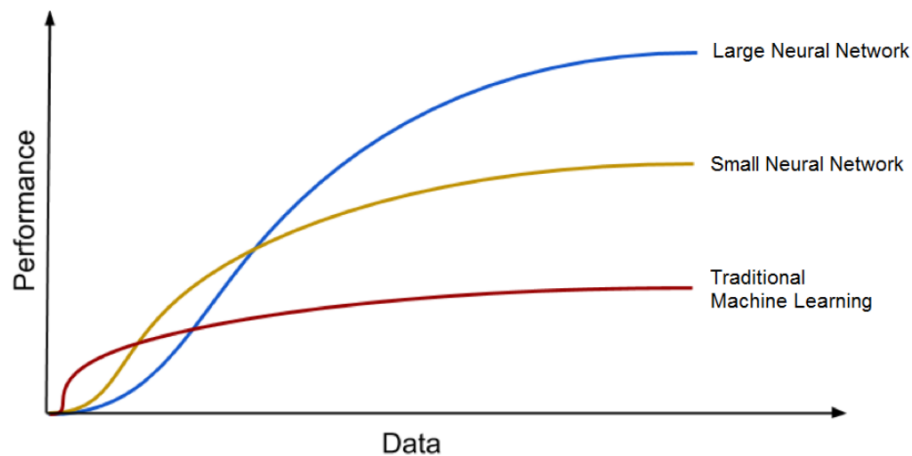


Figure 1.2 Deep learning algorithms improve as the volume of data increases (Oppermann, 2020)

While deep learning models' accuracy tends to rise with the amount of training data, traditional machine learning models like SVM and Naive Bayes Classifier stop increasing beyond a certain point (Oppermann, 2020). There are number of deep learning algorithm like RCNN, Faster RCNN, SSD and YOLO etc. Many researchers like such as (Redmon et al., 2016) did the comparison research on deep learning algorithms and found out that the YOLO is a fast, accurate object detector and also provides a significant performance. In this research, we are focusing to use YOLOv4 architecture to detect the classify pothole images.

Regions are used by algorithms like RCNN and Faster RCNN to locate the item inside the image. In case of You Only Look Once (YOLO) algorithm, examines the portions of the image which have a high possibility of presence an object rather than the entire picture. A single CNN layer in the YOLO algorithm forecasts the bounding boxes and their boxes class probabilities.

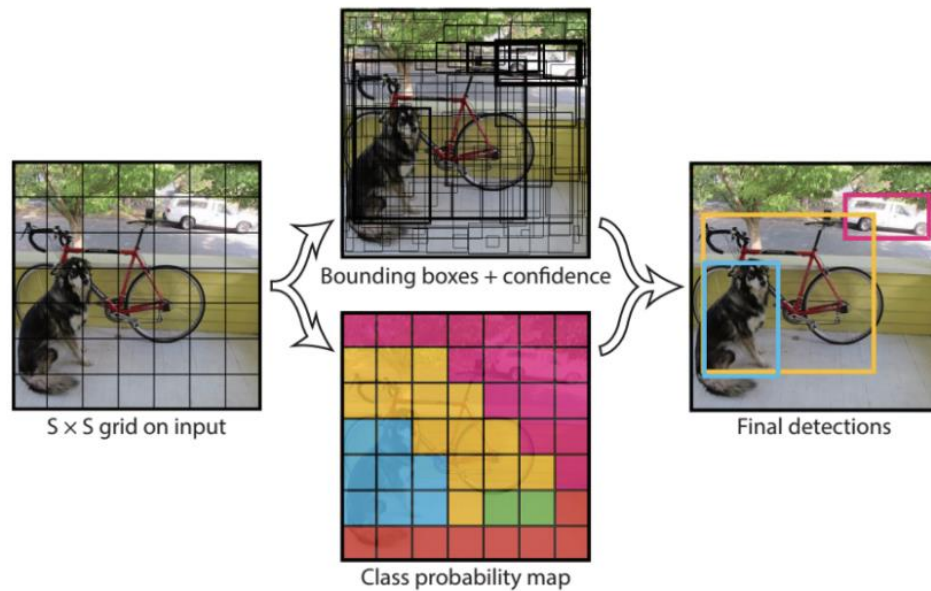


Figure 1.3 YOLO (Gandhi, 2016)

In YOLO, input image is split into $S \times S$ grid, each grid having m bounding boxes. This network creates a probability of class and an offset value for each bounding box. The

bounding box with their class probability value higher than a threshold value that is used to detect the object within the image (Gandhi, 2016).

1.1 Background

Road is one of the main sources of transportations for human and goods. Potholes are gradually increasing and expanding during the winter months, and these potholes can form very quickly. They are brought on by the groundwater's expansion and contraction after it has soaked into the pavement's subsurface. Major damages from potholes might include tire and wheel damage, damage to the chassis of a car, sudden breaking and slipping, accidents, and unfortunately death.

Potholes are one of the main primary reasons which directly affect road/street accidents and impact the daily normal routines of individuals. The most recent official statistics show that between 2018 and 2020, road accidents brought on by potholes resulted in 5,626 fatalities (Tribune News Service, 2022). Data from the Ministry of Road Transport and Highways (MoRTH) show that there were 2,015, 2,140, and 1,471 fatal road accident injuries in 2018, 2019, and 2020, respectively (Tribune News Service, 2022). Government always invest a selected money for road maintenance and recovery but at the end result of repairing road has not been effective due to improper management and care.

1.2 Problem Statement

In our daily lives, potholes are actually becoming a severe problem that requires expensive pavement and automobile repairs. The cars sustain substantial damage. Bent wheels, misaligned steering systems, punctured tires, damaged exhaust systems, and broken suspension in automobiles are a few notable examples. Road potholes play a major role in major accidents that happen all over the world. Road potholes, no matter how big or small, can result in accidents. When cars hit potholes, little accidents can sometimes turn severe. The impact of internal injuries and suffering can be severe and last a lifetime.

The selection of data representation or characteristics that machine learning algorithms are used to determine how well they function. Traditional methods for road damage detection based on shallow learning, such as PCA and SVM, have been confronted with issues such

angle variation, scene lighting, and the complexity of the image background. They only make use of a few fundamental properties of images and rely on simulation to obtain representative features. Deep learning-based techniques for pothole detection have become essential as a result of the realization of the aforementioned issues. The categorization and identification of photos with a single pothole can benefit from the use of deep learning models like CNNs. A variety of features can be extracted from images using CNN, which has many convolutional layers.

Pre-trained model has been using for pothole detection from past few years. In literature of pothole detection (Keval Doshi and Yasin Yilmaz, 2020; adak et al., 2021; park et al., 2021), it is found that pre-trained did not perform well, and also these models are overfitting due to the forgetting effect. To minimize the overfitting effect, models are trained with transfer learning and the configuration of hyperparameters.

Finding several potholes in a single image and accurately identifying them is the key challenge for pothole detection systems. This has motivated me to suggest a localization approach using bounding boxes, and the next step would be to determine whether the given image has a pothole, in this case, it would be identified. Alternatively, just label them as unknown.

1.3 Research Questions

RQ1: Which one of the YOLO variants is best suited for pothole detection?

RQ2: How can we use transfer learning and a hyperparameter tuning approach in the YOLO family to detect potholes?

1.4 Objectives

The following are the primary objectives of this thesis work:

- i. To localize and detection of potholes using the YOLO Family and compare with different performance matrices (RQ1)
- ii. To modify the existing YOLO architecture for the pothole detection model using transfer learning and improve the performance using hyperparameter tuning (RQ2)

1.5 Thesis Contributions

Our contributions to this research are summarized as follows:

1. Collected primary images of potholes from various regions of Nepal. We prepare the images set for YOLO format after data pre-processing and data augmentation.
2. To localize and detect potholes using different variants of the YOLO family and compare performance between them. The key models we compare in our research are YOLOv3, YOLOv3-tiny, YOLOv4, and YOLOv4-tiny.
3. Effectively utilizing the transfer learning approach to develop a pothole detection model that operates on our dataset. In the detection model instead of using the complete pre-trained model, YOLOv4 is customized to make it more suitable for training on a primary dataset for pothole detection.
4. We utilize the impact of state-of-the-art, Bag-of-Specials and Bag-of-Freebies approaches used for object detection throughout the pothole detection training.

1.6 Scope and Applications

In Nepal, potholes are a well-known issue and one of the primary reasons for serious accident issues. Since finding these potholes takes a long time, engineers are working on several machine learning and deep learning approaches to find them sooner rather than later so they can be fixed.

Since there are more vehicles on the road every year, we can take advantage of this by using sensors like cameras installed on cars to identify potholes and cracks because they provide varied results depending on the type of road.

The data produced by these sensors can be used to develop a classification model that will recognize the presence of cracks or potholes on the road. Deep learning methods can be more useful in this field rather than machine learning methods because deep learning models have several hidden layers, which are used to learn features by themselves. These statistics can be used by government officials to fill in those potholes, and the general public can get real-time updates on their progress.

This thesis study will help to improve pothole detection. With robust pothole detection and recognition, the use of such systems can be used in various sectors.

- Calculate road maintenance cost
- Travelling and maintaining the quality of roads.
- Government will have the facility of calculating the exact cost for road maintenance with knowing the exact potholes in roads.
- For travelers, this system helps to find the best road route for their destination.

1.7 Thesis Structure

This thesis details are defined as follows:

- Chapter 2
 - This section presents the brief literature survey that we found motivated conducting this pothole research.
- Chapter 3
 - This section explains the research methods which has been applied in detail. It includes our proposed model and details of experimental setup.
- Chapter 4
 - This chapter describes all the experiments and all the tools that are used in the thesis work in detail and also describes the results and discussion associated with all of those experiments of our thesis study.
- Chapter 5
 - This last section concludes the conclusion of thesis work showing the future possibilities of the research.

CHAPTER 2: LITERATURE REVIEW

Numerous researchers build a technique for examining highway/road properties with help of image capturing by cameras, in-vehicle cameras, and image processing methods to examine a road surface more efficiently. For instance, employing Naive Bayes-based machine learning techniques and image processing techniques, a study offered an automated method for detecting cracks in asphalt pavement (Chun et al., 2015). Additionally, a commercial black-box camera-based pothole-detecting system has previously been presented (Jo and Ryu, 2015).

Deep neural networks have recently made it feasible to analyze road surface damage fairly and precisely by Zhang et al. in (2017). For example, CrackNet, which calculate class scores for all pixels was introduced. These methods of identifying road damage only look for damage when it is present. However, changes in lighting and shadow in the road situation also result in lower detection performance. Although some studies classify damage according to type—for instance, Zalama et al. (2014) categorized pothole damage types horizontally and vertically, and Akarsu et al. (2016) classified damage into 3 types—vertical, horizontal, crocodile and the majority of research mainly concentrate on classifying damage into a few types.

In general, vehicles made especially for road inspection cost a lot of money. Recent years have seen substantial progress in smartphone sensor technology, and examples of road examination using one are becoming increasingly common. The use of mobile phones has the advantage of being able to check the road surface comprehensively at a very little cost. For example, (Buttlar and Islam, 2014) suggested a technique to determine the road flatness focusing the smartphone's accelerometer that is mounted in a vehicle. Moreover, (Casas-Avellaneda and Lopez-Parra, 2016) suggested a technique for visualizing potholes found by using sensors of smartphone (on a map). Furthermore, Mertz et al. (2014) suggested a way to detect damage of road surface with the laptop using road images collected by on

board cellphones mounted on vehicles that drive on a regular basis, such as trash trucks, normal bus, and passenger automobiles.

Using 2D photos provided by the Korean government, Seung et al. (2015) suggested a method for image-based pothole detection that offers improved accuracy and performance than the current detection approach. For bright photos as opposed to dark, shadowy images, the suggested approach performs and has superior accuracy. However, they failed to recognize shadow images and various shapes of potholes. Using the color distribution of road surface pixels, Ying et al. (2016) focused on the extraction of shadow-free features for real-time road damage identification.

Recent research has focused mostly on object detection with deep learning algorithms since deep learning approaches easily deal with feature extraction. (Young-Jin Cha and Wooram Choi, 2017) proposed a deep learning based road crack damage detection approach based on convolutional neural networks. In this research, they used a vision-based convolution neural network to investigate crack detection. They use stochastic gradient descent to train the network (SGD). The test findings revealed constant performance despite test photographs shot in a variety of lighting situations, including close-up, blur, shadow, and intense lighting. Additionally, the proposed method's performance was unaffected by working distance, camera specifications, or image quality.

Furthermore, Convolutional neural networks were used by Fan et al. (2018) to conduct research on automatically detecting pavement cracks using two separate datasets and to resolve the multi-label classification problem with unbalanced samples. A sigmoid function is used to activate a function for the result or output in a multi-label classification issue. Different state-of-art approaches are compared, the proposed method beats previous state-of-the-art approaches in terms of handling various pavement textures. Furthermore, (2018), Maeda et al. suggested a method for classifying and detecting road damage using deep learning networks and photos from smartphones. The goal of the research is to accurately categorize the eight different forms of harm that the Japanese government recognizes. The SSD MobileNet V2 and SSD Inception frameworks were used to train the model. The SSDs that use Inception V2 and MobileNet have relatively low memory and

CPU loads requirements while still maintaining high accuracy. However, a pixel-by-pixel approach are used to identify road/street damage and improve detection performance.

Numerous studies have also been conducted utilizing R-CNN to detect objects. In 2019, Amita Dhiman and Reinhard Klette presented an approach for locating potholes that makes use of a variety of algorithms, including YOLO, Mask R-CNN, Single-frame stereo vision, and Multi-frame Fusion-based methods. They discovered via their research that Mask R-CNN outperforms other techniques and algorithms in terms of accuracy and performance. Furthermore, Faster R-CNN with RPNs are main emphasis of Ren et al. in 2017 for real time for detection of object. The faster R-CNN build with methods. A deep fully convolutional network serves as the first module and recommends regions, while the Fast R-CNN detector serves as the second module and employs the suggested areas. Additionally, they concentrate on RPN and Fast R-CNN sharing features to reduce time usage and improve network efficiency.

By modifying the YOLOv2 architecture, Lim Kuoy Suong and Kwon Jangwoo (2018) conducted research on pothole identification. YOLOv2 performed well in terms of recall and precision, scoring 82.43% and 83.72%, respectively. Despite the scant learning pothole images and the difficult properties of pothole data, Ping et al. (2020) using deep learning techniques, suggest a pothole detection system that is effective and can identify potholes in the road automatically. With preprocessed data, four models - YOLO V3, Faster R-CNN, HOG with SVM and SSD are trained and tested where YOLOv3 gives 82% high accuracy among other models with 1500 images. A new 1500-picture dataset on Indian highways has been produced for this article by Dharneeshkar et al. (2020). YOLO is used to train the dataset and annotate it (You Only Look Once). The findings are compared after the newly from dataset has been trained on different YOLO variants. Tiny YOLOv3 achieved high mAP@0.50 with 49.71% and precision of 76% and recall of 40%.

Furthermore, (Ukhwah et al., 2019) suggested deep learning approach based on YOLO for asphalt pavement road pothole detection. YOLO predicts the entire image, the categorization, and the localization in a single step. Classes and bounding boxes are predicted using all picture features at once. There were three different Yolo v3 architectures used: Yolo v3-spp, Yolo v3-tiny and Yolo v3. The results of the research demonstrate that

the mAP of YOLOv3-SPP with 83.43%, YOLOv3-tiny with 79.33% and YOLOv3 with 88.93%.

Weighted-Residual-Connections (WRC), Cross-Stage-Partial-connections (CSP), C-IoU, Mosaic Data augmentation and Mish-activation were some of the novel features employed in 2020 by Bochkovski et al. They discovered the MS COCO dataset's 43.5%AP (65.7% AP50) at a real-time speed with 65 FPS using YOLOv4 model on Tesla V100. Using YOLOv4 model, Mohd Omar and Pradeep Kumar (2020) proposed a research article for pothole detection where its accuracy was evaluated based on recall, precision, and avg IOU which are 37%, 58%, and 38.38% respectively.

In (2020), Yasin Yilmaz and Keval Doshi conduct research on street damage detection using ensemble approach. With many different model versions, the ensemble approach was thoroughly tested, and it was successful in achieving an F1 score value with 0.628 on the first dataset and 0.6358 on the second dataset. Custom object detection and analysis in real-time using YOLOv4 with fine-tuning proposed by Adak et al. (2021). With 65.7% precision, YOLOv4 fine-tuning was successful. Park et al. (2021) researched on the application of different YOLO variants for computer vision based real-time pothole detection. Three cutting-edge object detection frameworks are tested, including YOLOv4-tiny, YOLOv5s, and YOLOv4 to evaluate how fine they perform on the basis of detection accuracy and real-time responsiveness. The result of the research demonstrated that YOLOv4 gives a better mAP with 78.7% and the mAP of YOLOv4 and YOLOv5s is 77.7%, and 74.8%, respectively. Pothole detection performance is compared using real-time Deep Learning algorithms (Al-Shaghouri et al., 2021) in a variety of settings, including SSDTensorFlow, YOLOv3-Darknet53, and YOLOv4-CSPDarknet53. In result, YOLOv4 achieved high precision of 85%, high recall of 81%, and 85.39% of mAP.

Nevertheless, overfitting is a significant issue when using deep learning. This is the situation where the model only performs well on training data and poorly on testing data; it has limited generalizability. Typically, this is because deep learning model training necessitates a huge amount of data. Additionally, it is expensive to generate additional data for each task. To address this issue, transfer learning has recently become popular. Pre-trained models like YOLO, ResNet50, VGG16 are used as examples. Transfer learning,

also known as the usage of a pre-trained model, is the process by which a model that has been trained for one job can be applied to another task; this is the idea of knowledge sharing. According to several studies, pre-trained models cannot function as predicted when the task domain varies, hence these models can be limited or adjusted to some extent to perform better (Li et al., 2018).

Due to various shadows and backgrounds, many researchers were unable to accurately locate potholes. Some designs, like the Faster RCNN architecture, solely prioritize speed above accuracy, and some studies merely do comparison analyses to determine whether the algorithm is more effective in detecting potholes. Studies that attempted to identify tiny potholes in an image likewise came up empty. When there are several potholes in a single image, many systems struggled with accuracy and performance. Our research aims to the use newly formed YOLOv4 algorithm and modify to localize multiple pothole objects and detect those potholes.

CHAPTER 3: RESEARCH METHODOLOGY

This section focusses on research idea and the methods that are used as a methodology for this research.

3.1 Research Method

The Following Steps were carried out to complete this research

- Data Collection
- Data Preprocessing
- Proposed Model
- Evaluation

3.1.1 Data Collection

Primary pothole images are gathered from various region of Nepal for the pothole detecting process. iPhone 13 pro, Samsung S8 plus and Samsung M32 mobiles phones are used to captured pothole images and images have different size and format according to the mobile phones. Initially 363 images were collected, after augmentation 865 images were obtained. The majority of the images were taken in Chitwan, Dhulikhel and Pokhara.



Figure 3.1 Primary pothole images

3.1.2 Data Preprocessing

Before pothole images are used for model training and inference, preprocessing is done to format them. This comprises, but is not restricted to, orienting, scaling, and adjustments of color. To expose the model to a broader range of training examples, image augmentation involves applying alterations to image to produce different versions of the same objects. For instance, randomly changing an input image's rotation, brightness, or scale necessitates that a model considers how the image subject can seem in multiple configurations.

3.1.3 Proposed Model

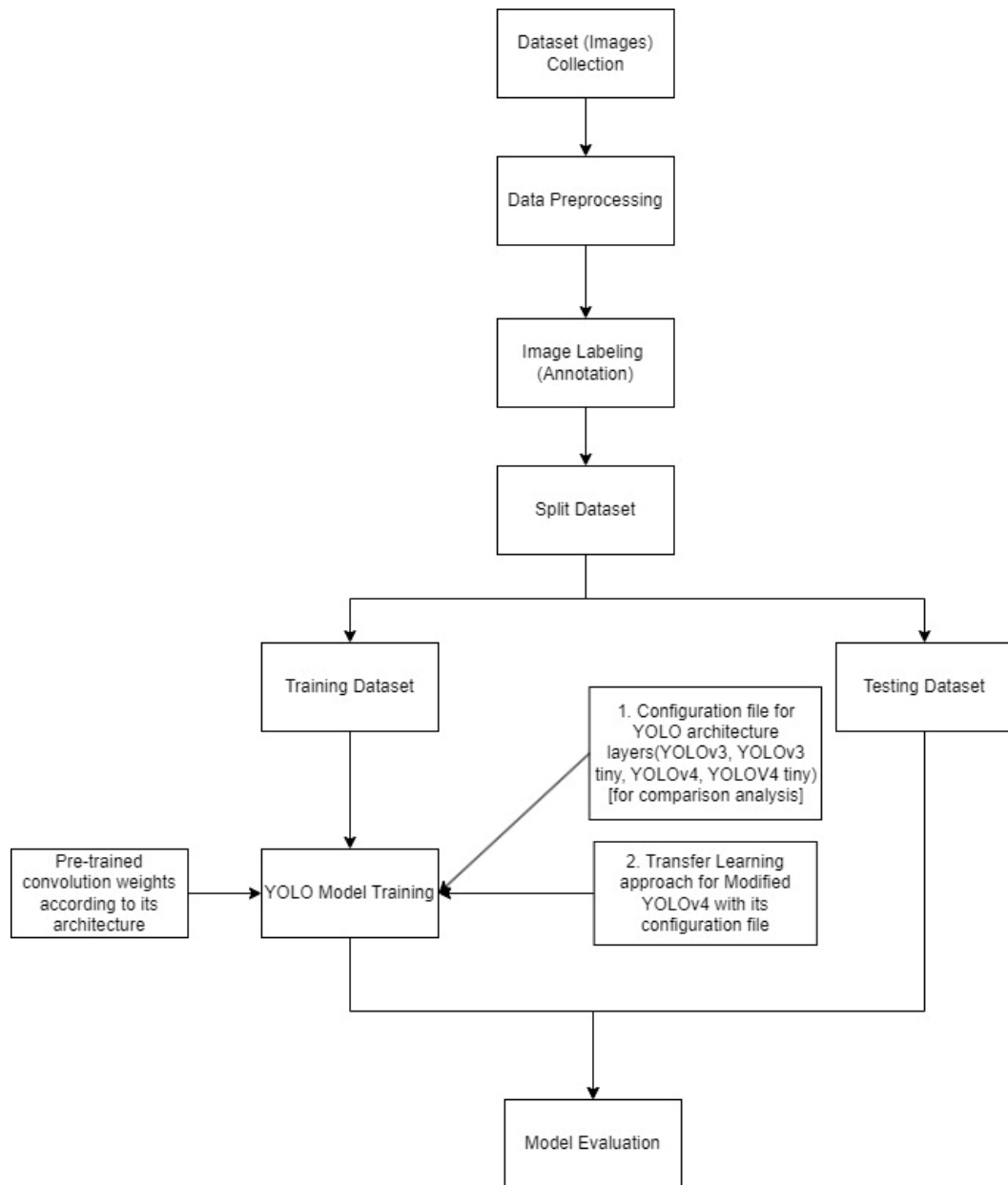


Figure 3.2 Proposed pothole detection model

In above figure shows that the general proposed model for YOLO family for model evaluation. After collecting images, images were annotated and then pre-processing was done to get proper format for YOLO model inputs. All the data are splits into training and testing parts for model training. Models are training with YOLO configuration file and pre-trained weights using transfer learning approach. At the end, models are evaluated using

confusion matrix, comparisons between different models and improved YOLOv4 models for pothole detections.

YOLOv3

The creation and improvement of YOLOv2 led to the creation of YOLOv3 (Redmon and Farhadi, 2018). The YOLOv3 approach has been shown to have a regression issue with object detection. YOLOv3 use logistic regression to calculate the confidence score for all presented bounding box. Small grid cells $S \times S$ are created from the input image using the YOLOv3 approach. An object must be detected by the grid cell if it enters a central cell. The bounding boxes' position information is estimated in each cell, and the objectivity scores of those boxes are computed as well. The confidence score must be 1 if the bounding box covers an object with known ground accuracy more than the other bounding boxes did (Zhao and Li, 2020).

Using logistic regression, how closely a bounding box covers an object is predicted by YOLOv3 (Redmon and Farhadi, 2018). Since the softmax layer didn't show to be very effective at improving performance, logistic classifiers are used. YOLOv3's performance has significantly improved when it comes to small objects, but when it comes to medium-sized and large objects, it still lags behind the outcomes from YOLOv2. YOLOv3 employs fifty-three convolutional layers. Bounding boxes are predicted using three distinct scales. An objectness score that is assigned to each bounding box. Multilabel classification is used to determine the class of the items contained within the bounding boxes. The last layer creates a 3D tensor with bounding boxes, objectness score, and prediction of class stored.

YOLOv3 Tiny

(Redmon et al., 2016) suggest another variant of YOLO v3 is referred to as YOLOv3-Tiny because the convolutional layer's depth has been reduced. Therefore, this result show that huge improvement in running speed (around 442% quicker than the previous YOLO versions), but a decrease in detecting accuracy. For feature extraction, the Darknet-53 architecture of YOLO v3 uses a number of 1×1 convolution layers in addition to 3×3

convolution layers (Xiao et al., 2019). In order to reduce the size for the convolution layer, YOLOv3-Tiny focuses on a pooling layer. An object objectness score, object bounding box, and predictions of class are all included in the three-dimensional tensor that is predicted at two different scales. It creates $S \times S$ grid cells out of an image. We won't use the bounding boxes whose objectness scores aren't optimal for final detections. The feed-forward configuration of YOLOv3-Tiny uses convolution layers and max-pooling layers for feature extraction of objects. Two alternative feature map sizes, 13×13 and 26×26 combined with an upsampled 13×13 feature map, are used for the prediction of bounding boxes (Ding et al., 2019).

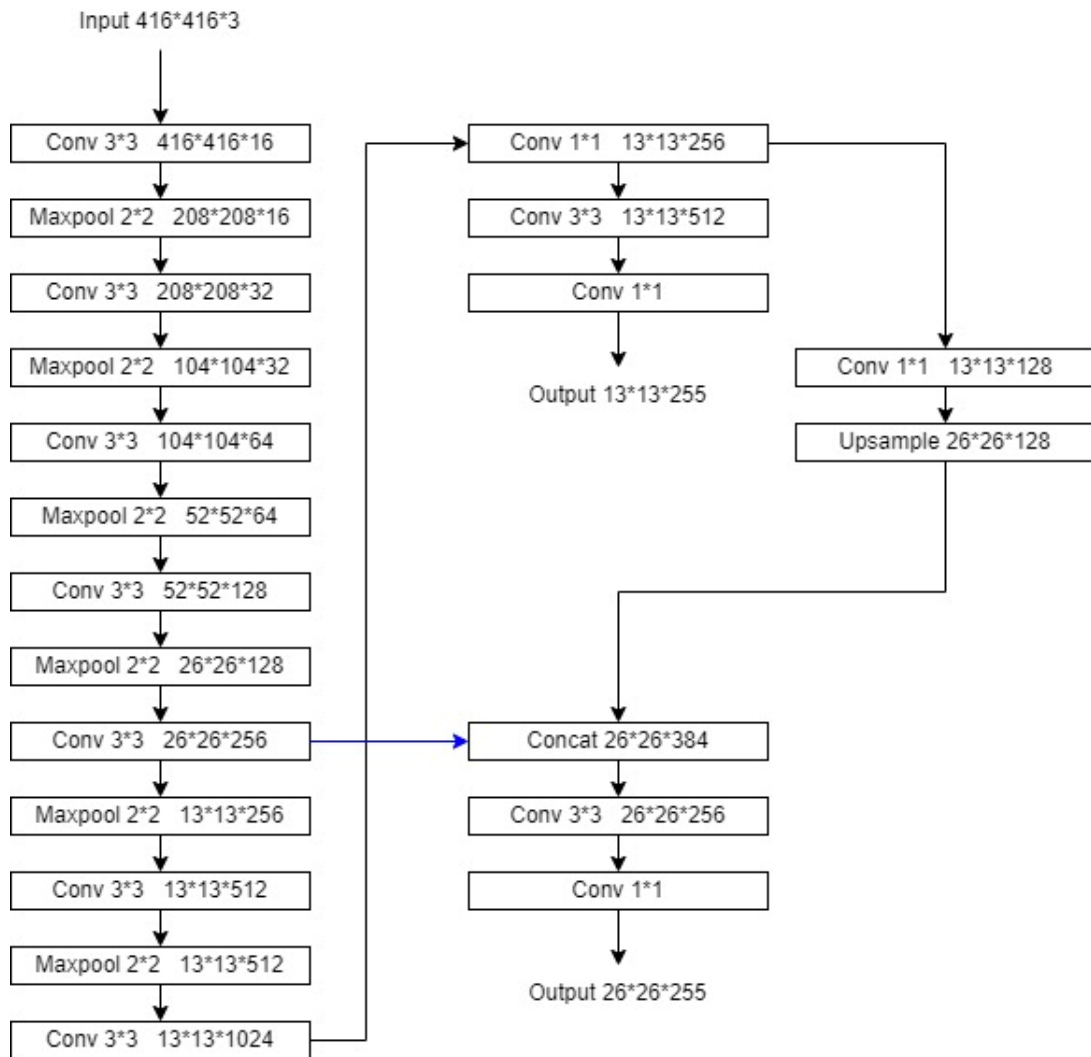


Figure 3.3 YOLOv3- Tiny architecture (Ding et al., 2019)

YOLOv4

The modeling process used in this research is YOLO architecture, Alexey Bochkovskiy and his team develop this architecture after adding some features in YOLOv3. Bag of Freebies and Bag of Species feature is used to enhance the performance of the new YOLO architecture. Some features are Cross Stage Partial Connections (CSP), Mish activation, Mosaic augmentation, Dropout regularization, and CIoU loss and combine a few of features to get best outcomes.

One stage Detector

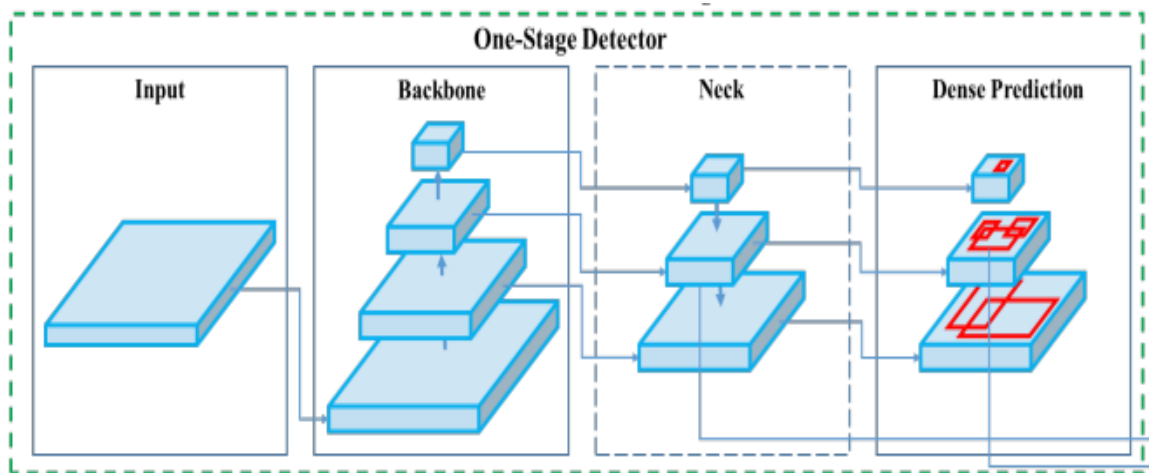


Figure 3.4 YOLOv4 architecture (Bochkovskiy et al., 2020)

Complete the architecture, needs four main parts;

- Input
- Backbone
- Neck
- Dense Prediction

Input

Input is the initial step in this architecture. In our case pothole images is the input object for this research.

Backbone

In YOLOv4 architecture, the feature extraction method can be done in a backbone part. We can use YOLO-tiny, Darknet53, and CSPDarknet53 as feature extraction methods. In this research, we have used CSPDarknet53.

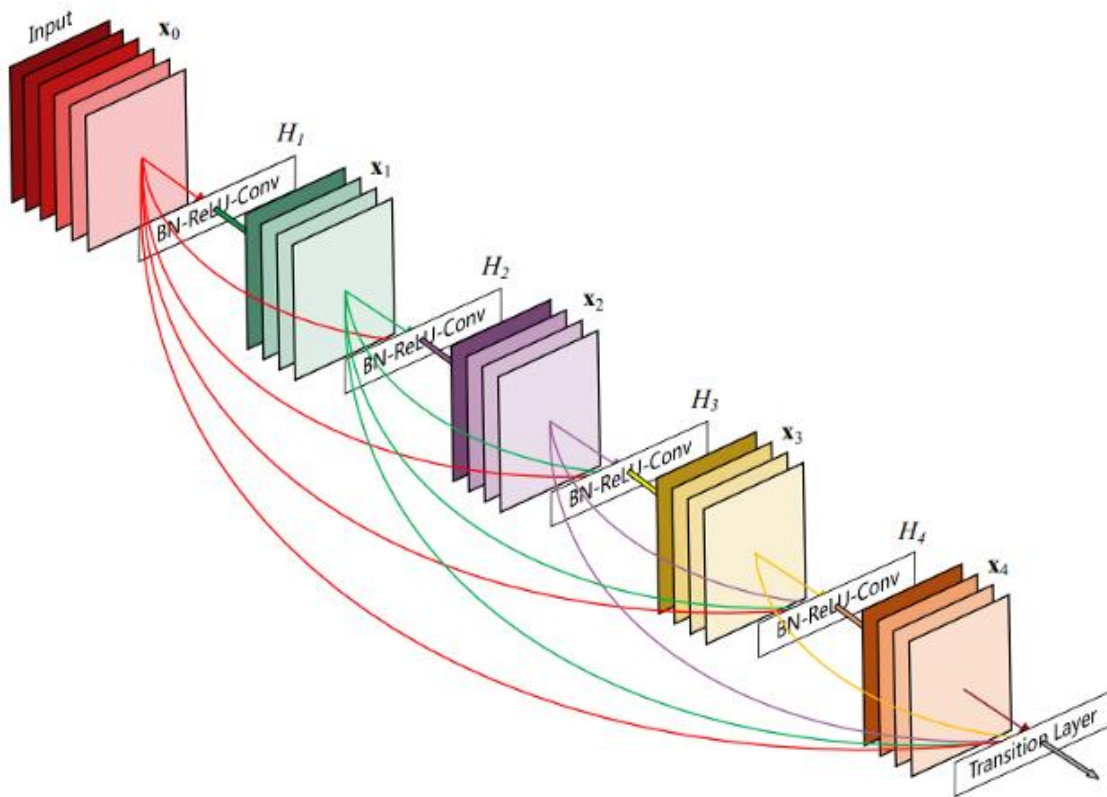


Figure 3.5 A dense 5-layer block. Each layer inputs all previous feature-maps. (Huang et al., 2017)

Each layer H_i in the YOLOv4 backbone's dense block underwent numerous convolution layers, with batch normalization (BN), ReLU, and convolution coming after each layer. In

YOLOv4, Hi uses the output of all preceding levels as well as the original as input rather than only the output of the last layer, i.e. $x_0, x_1, \dots$, and x_{i-1} .

Four feature maps are output by each Hi above. As a result, the growth rate of the number of feature maps at each layer is raised by four (Huang et al., 2017).

In CSPDarknet53, CSP means Cross Stage Partial which splits the given input feature map of the DenseBlock into two different processes. first process will go through the convolution and other processes won't. After these two processes, CSPDarknet aggregate the result (Wang et al., 2020).

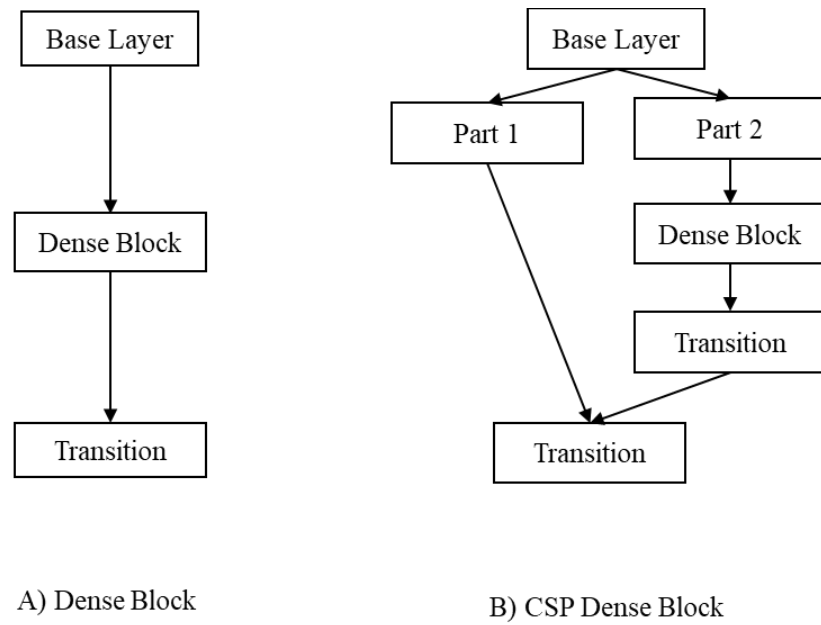


Figure 3.6 DenseNet and CSPDenseNet (Wang et al., 2020)

The DenseNet design, which concatenates the prior input and recent input information before proceeding into dense layer, is where cross stage partial architecture originates.

Each stage layer in DenseNet has a transition layer, a dense block, and k dense layers that make up each dense block. Concatenating the output and input of the i^{th} number dense layer

will result in the output becoming the $(i + 1)^{th}$ dense layer's input (RUGERY, 2020). The equations illustrating the above procedure can be written as follows:

$$\begin{aligned}
X_1 &= W_1 * X_0 \\
X_2 &= W_2 * [X_0, X_1] \\
&\vdots \\
&\vdots \\
&\vdots \\
X_k &= W_k * [X_0, X_1, \dots, X_{k-1}]
\end{aligned} \tag{1}$$

In the equation above, $*$ stands for the convolution operator where $[X_0, X_1, \dots]$ indicates to concatenate X_0 , X_1 , ..., and X_i , and W_i stands for the i^{th} dense layer's output and weights, respectively.

The CSP and denseNet are both built on the same theoretical foundation; however, the author separated the input k^{th} into two parts, x_0'' and x_0' , so that one portion will go through the dense layer and the other half, x_0' , will be concatenated with the output of the dense layer at the end (RUGERY, 2020).

$$\begin{aligned}
X_k &= W_k * [X_0'', X_1, \dots, X_{k-1}] \\
X_T &= W_T * [X_0'', X_1, \dots, X_k] \\
X_U &= W_U * [X_0', X_T]
\end{aligned} \tag{2}$$

Neck

The next step is the neck which adds an extra layer in between the the head (dense prediction block) and backbone. It gathered feature maps made at various stages. A neck is typically made up of numerous top-down and bottom-up routes. SPP and PANet are the best methods for YOLOv4 architecture.

For different size of feature maps, SPP will generate same fixed size of features. To create a fixed feature map size, SPP can be used for pooling layers, for example; Max Pooling and generate various representations of our feature maps.

In PANet, Path Aggregation Network used to aggregate the information to get improved accuracy. In the YOLOv3, a newly formed vector can create by combining data from the current layer and preceding layer. In the YOLOv4 implementation, a new feature is more focused, where the newly formed vector is generated by concatenating the previous input and new input value.

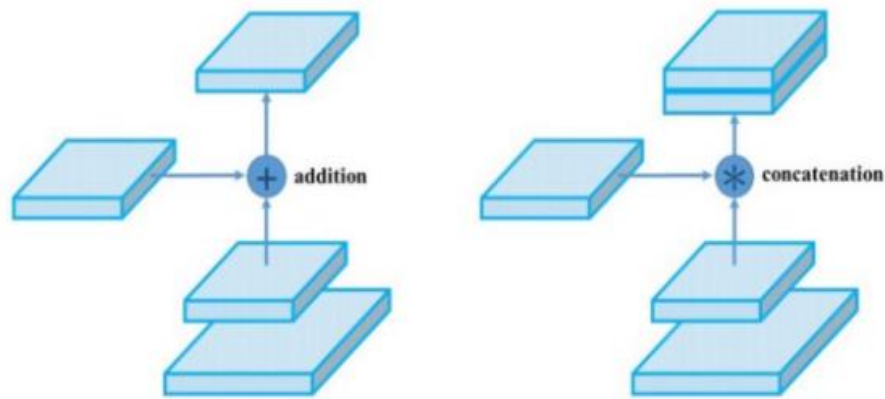


Figure 3.7 PAN and YOLOv4 PAN (Bochkovskiy et al., 2020)

When the SPP block is add to the YOLOv4 design after the CSPDarknet53, the receptive field is greatly increased, the most important context features are divided, and the network operation performance is not as significantly slowed down. We substitute PANet for YOLOv3's FPN as a method of parameter accumulation from various backbone stages for various detection levels defined by Bochkovskiy et al. in 2020.

Head (Dense Prediction)

Dense prediction is the principal function of the head portion in a one-stage detector. The bounding box coordinates for a class (x, y, w, h) and the confidence score are both detected by the head network. To predict the probability that an object will appear in each cell using anchor boxes, the YOLO principle first splits the photos into a number of grid cells. The

predicted result is a vector with its probability classes and bounding box coordinates (Lessons, 2020).

In this study, we select PANet as a path-aggregation neck, SPP as an extra module, CSPDarknet53 as the backbone, and YOLOv4 architecture has a YOLOv3 anchor based head.

Selection of Bag of Freebies (BoF) and Bag of Specials (BoS)

The Bag of Freebies and Bag of Specials are the added method in YOLOv4 compare to YOLOv3. These methods are used in object detection training to get better results.

Bag of Freebies (BoF)

The bag of freebies process is the collection of processes that only rise the training cost or modification in the training method while leaving the low inference cost. The training process can be improved in specific ways that boost accuracy without slowing down inference (e.g., cost function, data augmentation, soft labeling, class imbalance).

For instance, data augmentation satisfies the requirements of the Bag of Freebies definition. Data augmentation raises the input object's variability so that the created object detection method is more robust to images from diverse environments. The methods MixUp, CutMix, Photometric Distortion, and Geometric Distortion are the most frequently employed in research.

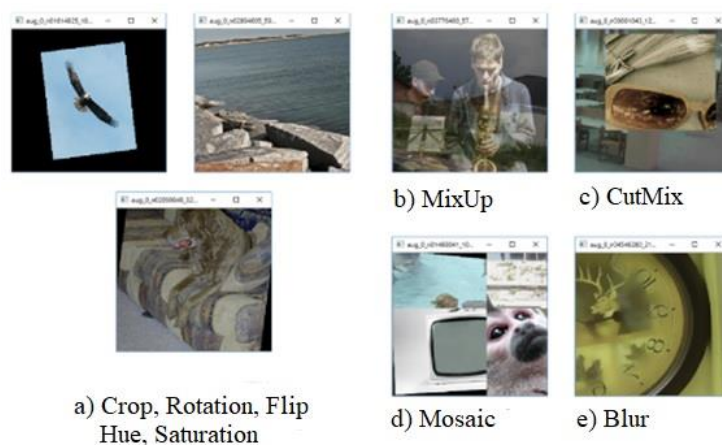


Figure 3.8 Data augmentation technique (Bochkovskiy et al., 2020)

1. Mosaic data augmentation

It is one of the most used data augmentations in newly research era, which is also used in our research paper. Mosaic data augmentation joins four training images into single image in certain ratios. The strategy allows the model to acquire the ability to detect the objects that are smaller in scale than usual. Additionally, it helps the model localize different kinds of objects in various parts of the image frame (Bochkovski et al., 2020).



Figure 3.9 Mosaic data augmentation (Bochkovski et al., 2020)

2. CIOU Loss (Complete-IoU)

To find the bounding box loss, CIOU loss is a newly formed loss calculation compared to the IoU loss. It provides the two key ideas. The first method is the central point distance, which is defined as the distance between the expected and real centers of the bounding boxes. The aspect ratio of the real bounding box and the expected bounding box are compared in the second method.

$$L_{CIOU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (3)$$

where b and b^{gt} stand for the mid points of B and B_{gt} (ground truth), $\rho(\cdot)$ represents the Euclidean distance, c represents the diagonal length of the minutest enclosing box covering the two bounding boxes, α represents the positive trade-off parameter, and v for the consistency of aspect ratio detection (RUGERY, 2020).

$$v = \frac{4}{\pi^2} (\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h})^2 \quad (4)$$

where w represents the bounding box width and h represents the bounding box height.

3. Eliminate grid sensitivity

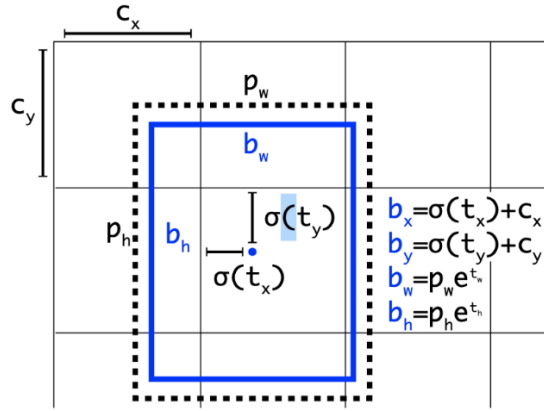


Figure 3.10 Bounding box with equation (RUGERY, 2020)

In YOLOv3, the grid equation is $b_x = \sigma(t_x) + c_x$, $b_y = \sigma(t_y) + c_y$, where c_x and c_y are real ways whole numbers for evaluation of the object coordinates, in this scenario t_x need extremely high value for b_x to equal with the value of c_x or $c_x + 1$ (RUGERY, 2020). In YOLOv4, writers find a solution by multiply by the sigmoid value by a factor exceeding 1.0, which gets rid of the undetected item effect on the grid.

Bag of Specials (BoS)

Bag of Specials improves the accuracy of object detection using plug modules and post-processing technique but it increases the inference cost by little. These post-processing techniques for screening model prediction results and plug-in modules are for adding some

attributes in models like attentions mechanism, strengthening feature integration capability, enlarging receptive field, etc (Bochkovskiy et al., 2020).

1. Mish activation

Mish is a smooth, non-monotonic activation function, unbounded above, bounded below property which help to increase performance compared to ReLU and other activation function which can be formulated as:

$$f(x) = x \times \tanh(\zeta(x)) \quad (5)$$

where $\zeta(x) = \ln(1+e^X)$, is a SoftMax activation function.

Unbounded above property used to speed the training process by avoiding the saturation process which causes the training to slow down drastically. The bounded below helps to achieve strong regularization effects. Non-monotonic functions are used to save the small negative value to stabilize the network gradient flow while the most commonly used ReLU failed to preserve a small negative value. Mish activation function is costly as compared to other activation functions but it gives better result in deep neural networks (R, 2020).

YOLOv4-Tiny

As a condensed form of Yolov4-Tiny, it is simple to run on mobile terminals or device terminals due to its tiny processing requirements, simple network structure, and low computing load. YOLOv4-tiny can be used for quicker training and quicker detection. The head, neck, and backbone networks make up the Yolov4-Tiny network architecture.

The primary architecture of Yolov4-Tiny implements the CSPDarknet53-tiny network, which primarily consists of two conventional convolution approaches and three CSP structures. CBL structure, which consists of the convolution layer, BatchNorm layer, and LeakyRelu, is adopted by the common convolution process. Yolov4-Tiny uses an FPN structure in the Neck section, integrating features from various scales to apply the deep network's rich semantic information and shallow network's geometrical details to improve feature extraction. To produce the final forecast, Yolo Head employed the feature map that

FPN had gathered. This resulted in two prediction scales, measuring 26×26 and 13×13 , respectively.

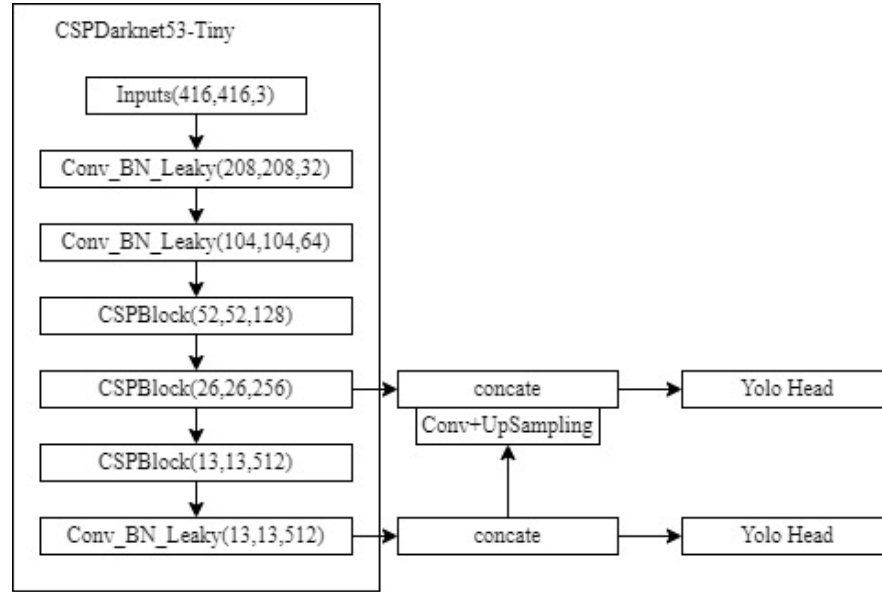


Figure 3.11 YOLOv4-Tiny architecture (Liu et al., 2022)

3.1.4 Evaluation Framework

A pothole classification model's effectiveness on a set of test datasets for which the true values are known is often described using a confusion matrix. The total number of accurate and inaccurate predictions in each class are combined with count values and grouped into a number of categories. Two expected classes could exist (yes) and (no). In our case, we are predicting the presence of a pothole, for example, (yes) means they have the pothole in the image, and (no) means they don't have the pothole in the image.

A confusion matrix framework was utilized to validate the model and evaluate its performance. Accuracy, recall, precision, F1-score and precision are calculated using the confusion matrix. The output of the confusion matrix is a matrix that details the entire performance of the model.

Table 3.1 Binary classification confusion matrix

	Predicted True	Predicted False
Actual True	TP	FN
Actual False	FP	TN

Confusion Matrix Parameters for model evaluation are as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (6)$$

$$\text{Precision(Exactness)} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (7)$$

$$\text{Recall(Completeness)} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (8)$$

$$\text{F1 score} = 2 \times \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

Intersection Over Union

Intersection area and union area of the actual and predicted bounding boxes are measured in terms of IoU, which is also known as the IoU ratio (Drid et al., 2020). The ground truth box is shown in green in Figure 3 below, while the model's prediction is shown in red. These two boxes' coordinates clearly differ from one another. The intersection point between two boxes is known as the intersection area, and the union area refers to the entire region enclosed by both bounding boxes (Li et al., 2020).

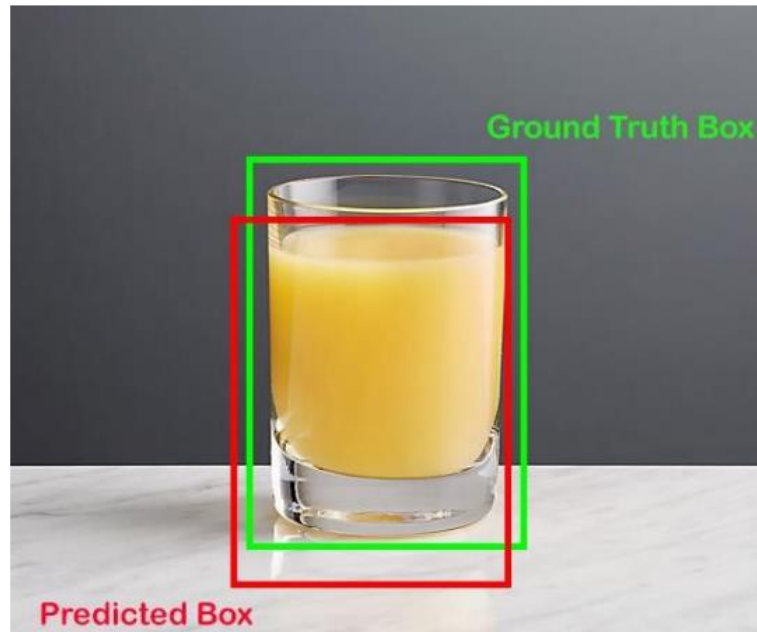


Figure 3.12 IoU and ground truth box and predicted box (Drid et al., 2020)

The possibility that an object is present in an anchor box can be used to define the confidence score. A classifier predicts the confidence. A projected box can be classified as a true negative, false positive, true positive, or false negative using both the IoU and confidence. For the IoU, a threshold value of 0.5 is predefined (Drid et al., 2020).

CHAPTER 4: EXPERIMENTS AND RESULTS

In this section, we give a brief clarification of tools used and the experiment which is conducted to fulfill the research objective in the below area.

4.1 Experimental Setup

4.1.1 Tools Used

The darknet framework used to build the pothole modeling process was written in C and CUDA, and it complies through Google Collab on a chrome browser with the following laptop specs: Intel (R) Core (TM) i7-8750H CPU @ 2.20GHz, 16 GB Memory, Windows 10 operating system, and Google Colab with a single 12GB NVIDIA Tesla K80 GPU.

4.1.2 Dataset

In the field of pothole detection on the highway, road and street surface, images of potholes are clicked from different areas of Nepal like Chitwan, Dhulikhel, and Pokhara. A total of 363 pothole images are captured from different mobile phones with different camera pixels. The images were taken from a variety of distances, angles, and lighting situations. The images were also captured under a variety of climatic situations (clear, overcast, rainy, etc.). The image quality is taken into consideration because deep learning relies heavily on it. 416×416 was the new size for the photos.

Table 4.1 Pothole images details

S. N	District	No. of Images	Places
1	Chitwan	181	Bharatpur, Tadi, Padampur, Manakamana-Muglin road
2	Dhulikhel	82	28 kilo, Panauti, Kathmandu University area
3	Pokhara	100	Sarangkot, Sera

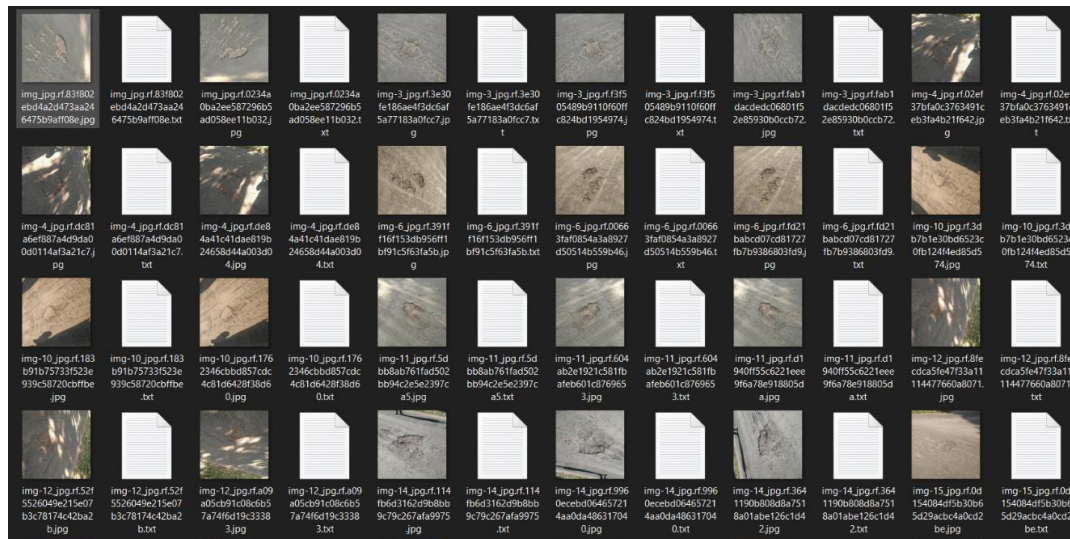


Figure 4.1 Train image

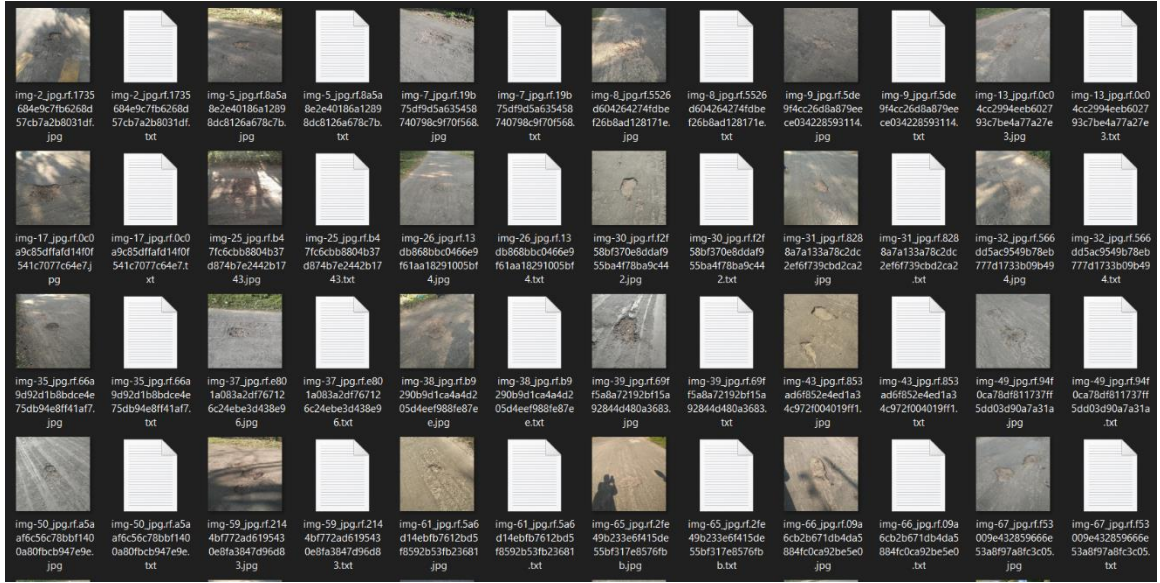


Figure 4.2 Test image

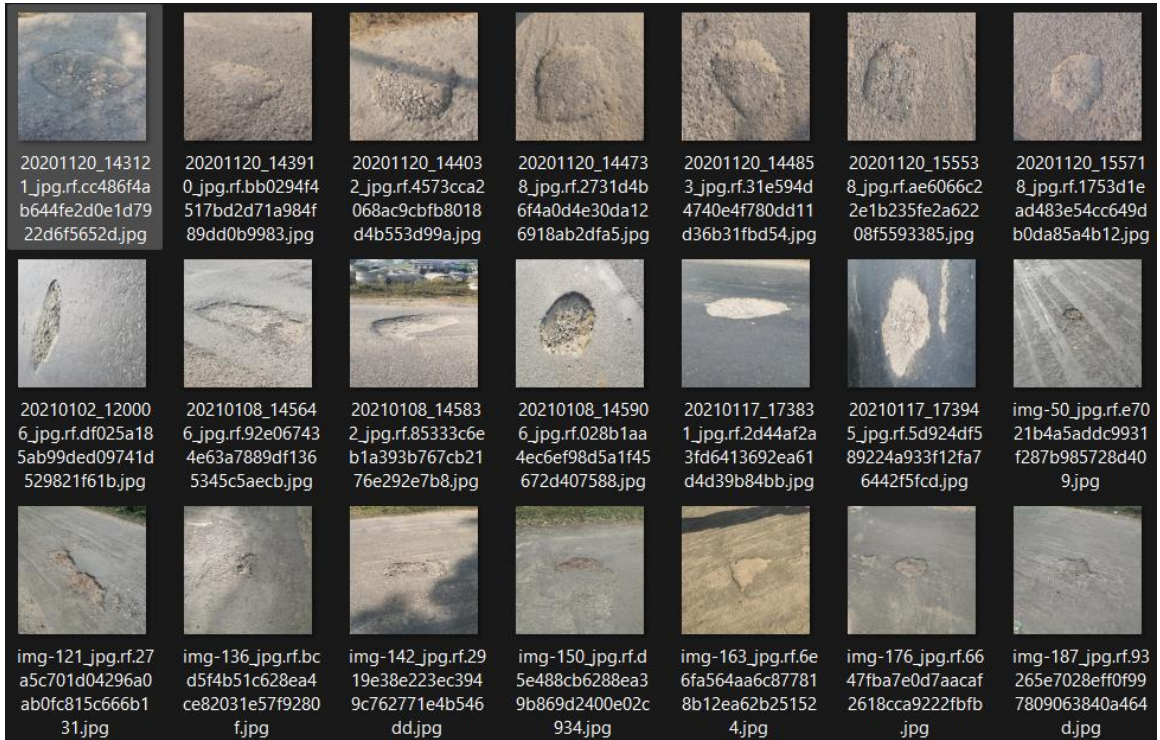


Figure 4.3 Test 2 image

4.1.3 Data Preprocessing

Initially, image augmentation is performed to expand images for better training and to reduce overfitting problems. To build a large dataset, A horizontal flip and rotations at a -45/+45 -degree angle are used (Stivaktakis et al., 2019). After augmentation 865 images were created from 363 images.

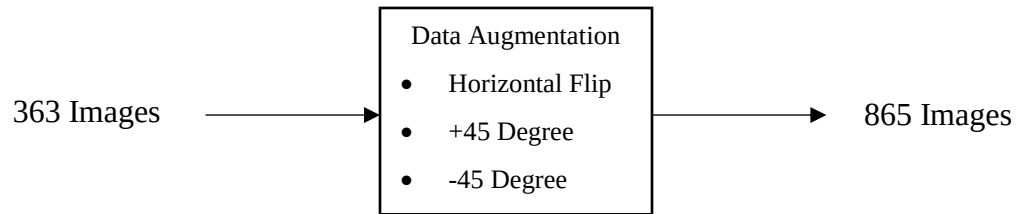


Figure 4.4 Data augmentation

Network architecture and input data type must be carefully considered when developing a successful neural network model. Image annotation and labeling process to generate a bounding box in images. This process can be done by the labeling tool to create an input format for different YOLO versions. Image enhancement may be necessary before feeding the images for training. The most common parameters are the image height, image width, number of images and the number of channels. As CNN requires all of the input pothole images to be of the fixed size, image resizing is to be done. Images are resized into 416×416 format for input.

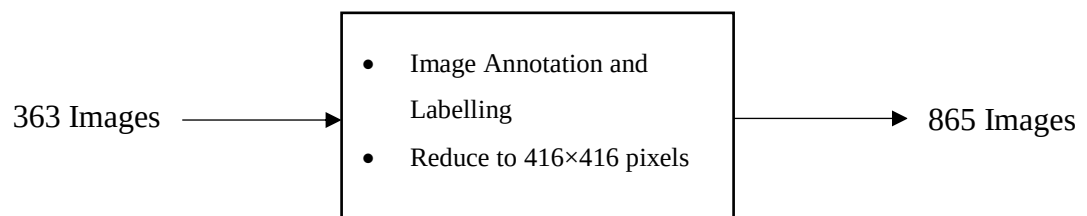


Figure 4.5 Create YOLO format images

The image information for the labeling and annotation process is class_id, width of bounding box (w) and height of bounding box (h) and x and y are bounding box center.

The value of class_id is in integer form and starts with zero and the value of the bounding box is between zero and one. Every pothole image is in the .jpg extension and a file is in .txt extension with pothole bounding box information. Samples of labeling and annotation are shown in the figure below.

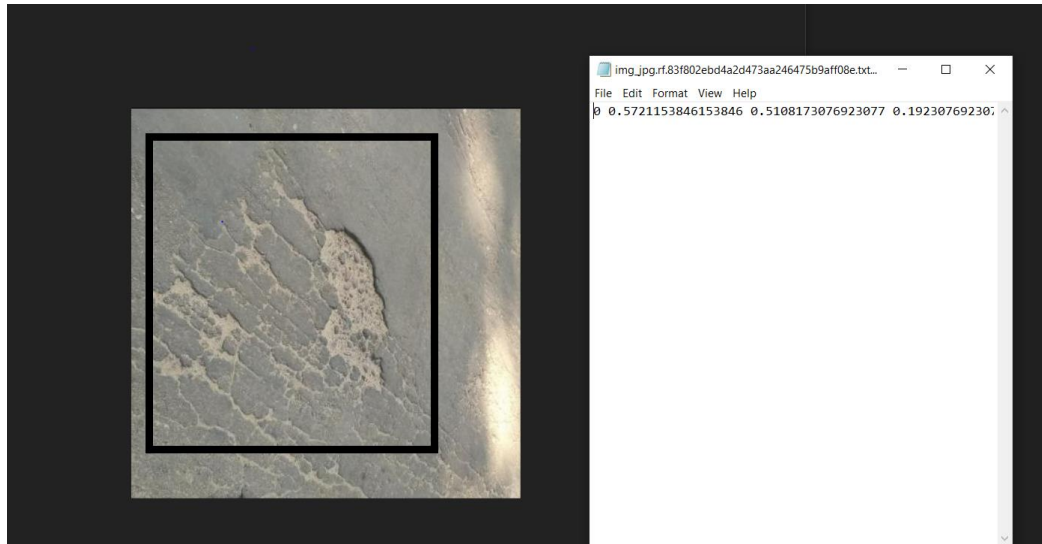


Figure 4.6 Annotated image

4.2 Experiments

This thesis paper's main propose is to improve the pothole detection performance using different YOLO variants. We have collected a pothole image from different devices from different places and did a preprocessing and data augmentation part for the proper input image for YOLO variants.

In experiment 1, different train/test ratios were tested to find the best ratio for further comparison of different YOLO variants in experiment 2. In experiment 3, the transfer learning approach were used with hyperparameter tuning to improve the performance of YOLOv4 and compare its result with other similar pothole detection research in experiment 4.

For performance evaluation, mAP, F1-score, Recall, Precision, and Avg IoU are calculated. The overlap between our prediction and the actual is measured by the intersection over the

union. The confidence threshold is maintained at 0.25. Additionally, the IoU threshold of 0.50 is used to calculate the mean average precision. As result, mAP is one of the critical indicators in evaluation.

4.2.1 Experiment 1: Using Different Train/Test Ratios

For these experiment images were randomly shuffled into a training and testing dataset with 3 different ratios and additionally random images were collected to test the performance of pothole detection system. The labeled train dataset is used to train the different variants of the YOLO family. The output of this modeling phase is a weight or also called the model. To calculate the model performance, area measurement (Intersection Over Union) and detection with the testing data are used.

Table 4.2 Performance evaluation for different train/test ratios

Train: Test Ratio	Avg IoU	Precision	Recall	F1-score	mAP
87:13	54.43	0.76	0.86	0.81	83.44%
80:20	59.49	0.81	0.84	0.82	82.84%
70:30	61.31	0.82	0.82	0.82	82.09%

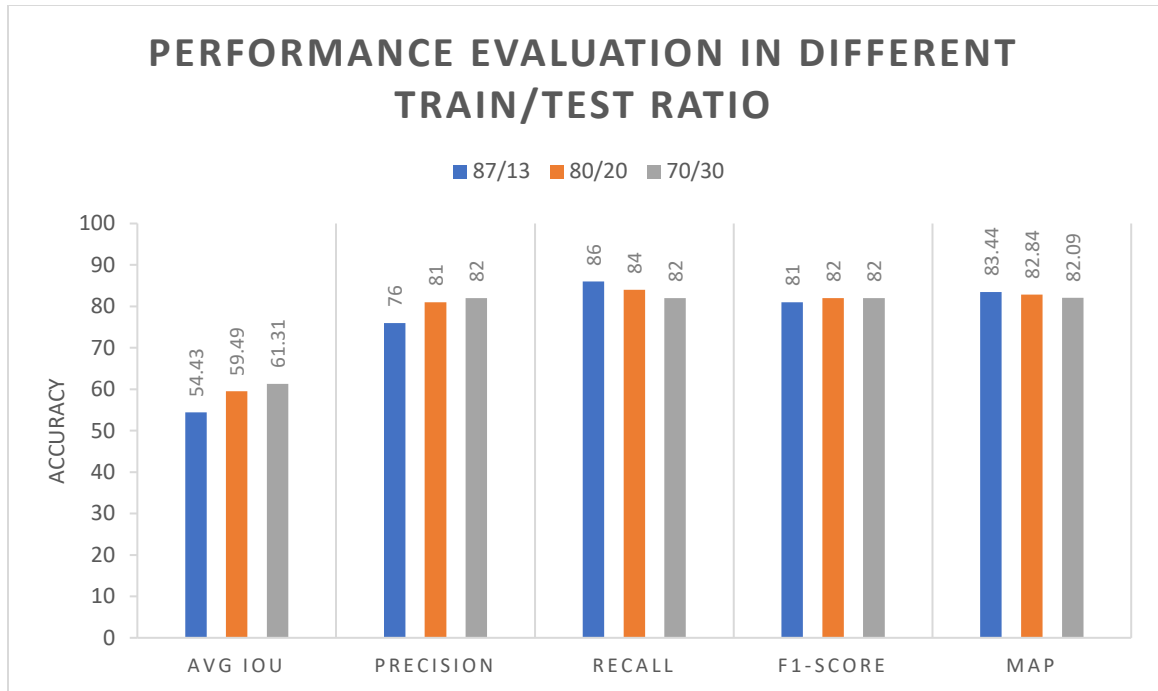


Figure 4.7 Performance evaluation in different train/test ratios

To evaluate the best train/test ratio for various YOLO models in the case of pothole detection, 87:13, 80:20 and 70:30 splitting ratios are used. In above Table 4.2 and Figure 4.7, the 87/13 split ratio has a higher recall value of 0.86 which means, it has a high value of successfully predicted corrected positive pothole from total actual positive potholes with compare to other splits and also a high mean average precision value with 83.44%. For intersection over union, the 70/30 splits ratio has a better value of 61.31 and also a high precision and f1-score with the same value of 0.82.

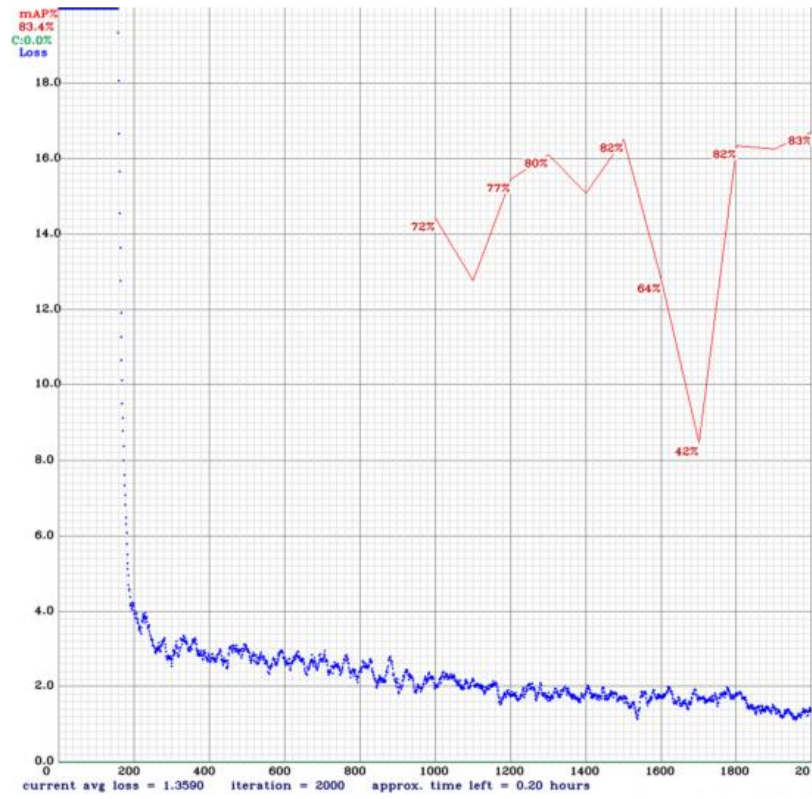


Figure 4.8 Mean average precision and IoU loss in 87/13 ratio

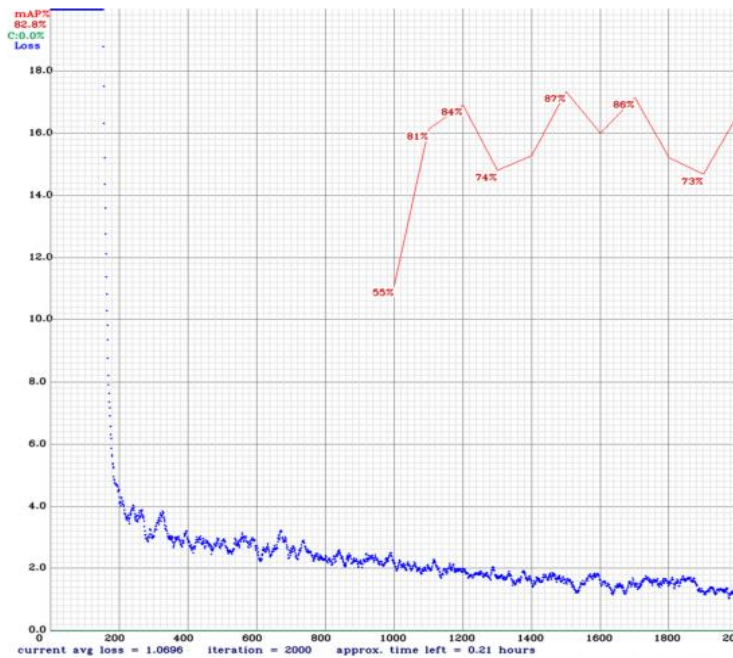


Figure 4.9 Mean average precision and IoU loss in 80/20 ratio

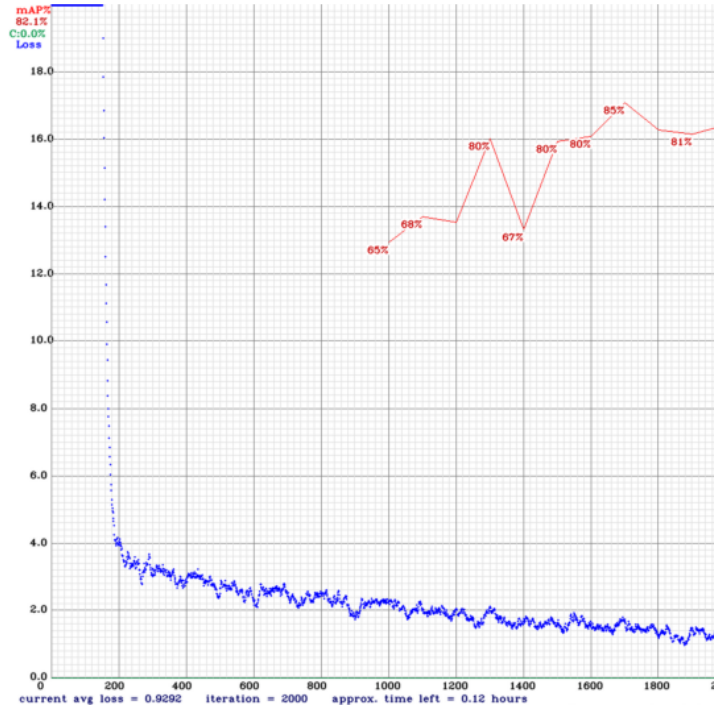


Figure 4.10 Mean average precision and IoU loss in 70/30 ratio

The error on the training dataset is represented by the blue curve as training loss. CIoU loss is used for calculating IoU loss. IoU loss gradually decreases with the increasing number of iterations as shown in the above figure.

The red line represents the mAP at 0.50 intersection over the union threshold. This determines if this model generalizes successfully on a previously unseen dataset or validation set. Mean average precision shows different variation in value because of different mini-batches but gradually increase its value concerning the number of iterations.

4.2.2 Experiment 2: Comparative Analysis of the YOLO Family

In this experiment, four different variants (YOLOv3, YOLOv3-Tiny, YOLOv4, YOLOv4-Tiny) of the YOLO family were compared for performance analysis. For comparison of YOLO variants, primary images of potholes were used.

Table 4.3 Performance evaluation of YOLO family

Models	Avg IoU	Precision	Recall	F1-score	mAP
Yolo-v4	54.43	0.76	0.86	0.81	83.44%
Yolo v4-tiny	47.85	0.70	0.82	0.75	73.48%
Yolo v3	58.89	0.85	0.82	0.83	82.06%
Yolo v3- tiny	34.29	0.52	0.49	0.51	40.27%

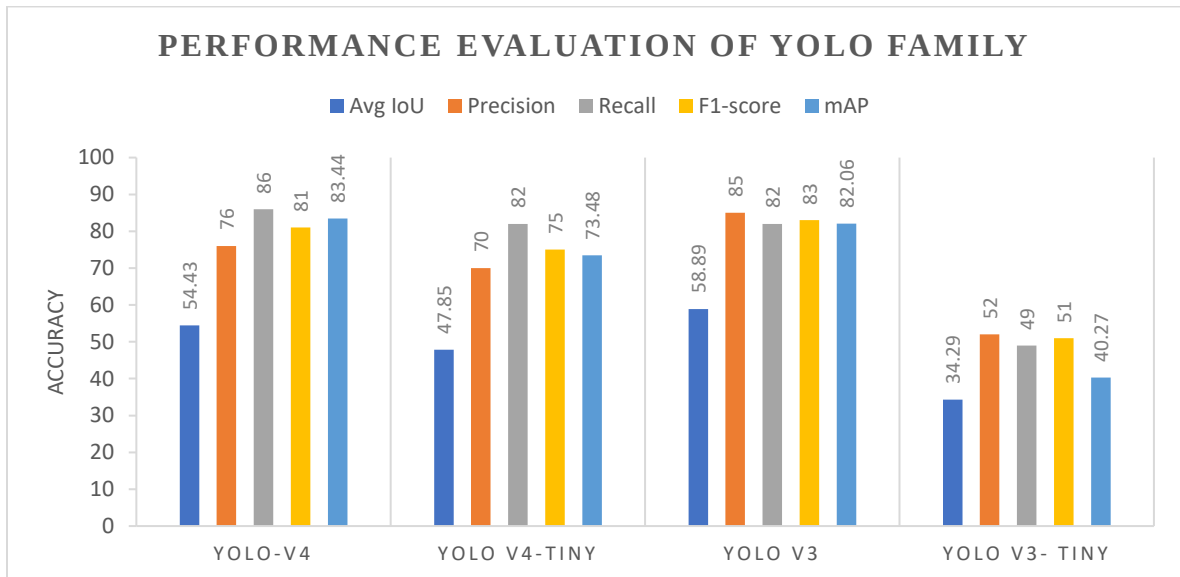


Figure 4.11 Performance evaluation of YOLO family

YOLOv3, YOLOv4, YOLOv4 tiny and YOLOv3 tiny are performed for comparative analysis. Table 4.3 and Figure 4.11 above demonstrate that the YOLOv4 model performs better than other YOLO models, with a mAP of 83.44% and recall of 0.86. YOLOv3 also performs well in terms of precision, f1score, and average IoU with 0.85, 0.83, and 58.89 respectively. Among all the models, YOLOv3 tiny performed poorly.

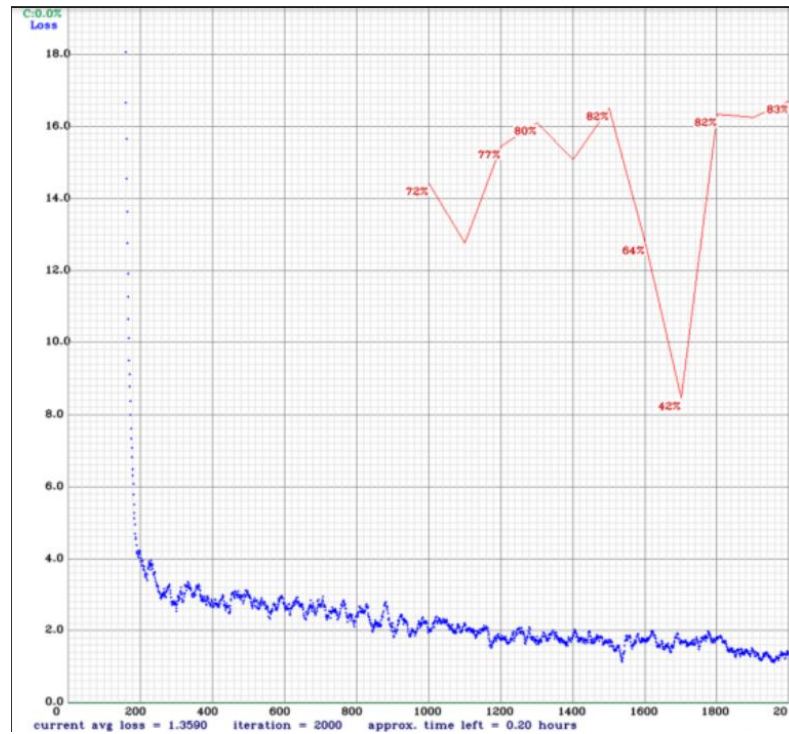


Figure 4.12 Mean average precision and IoU loss of YOLOv4

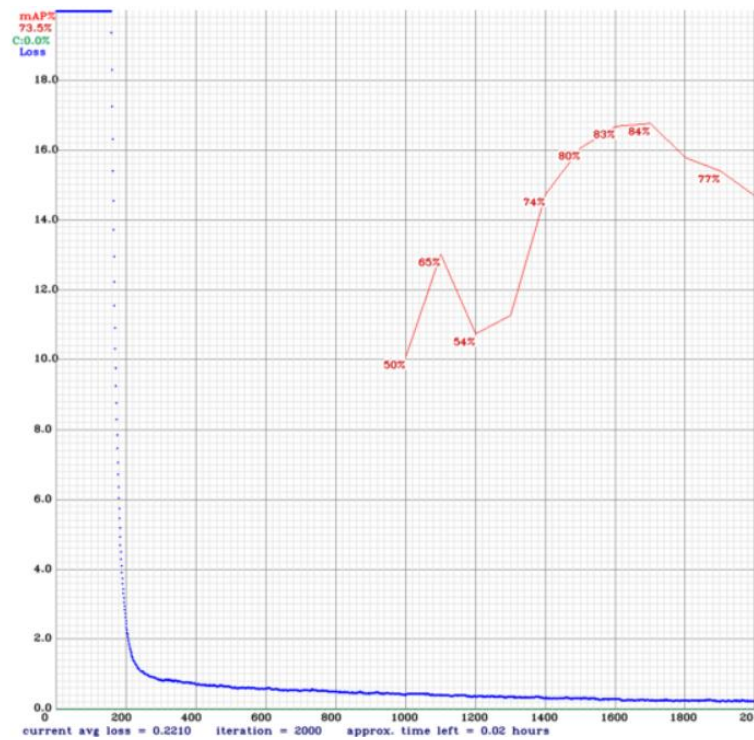


Figure 4.13 Mean average precision and IoU loss of YOLOv4-tiny

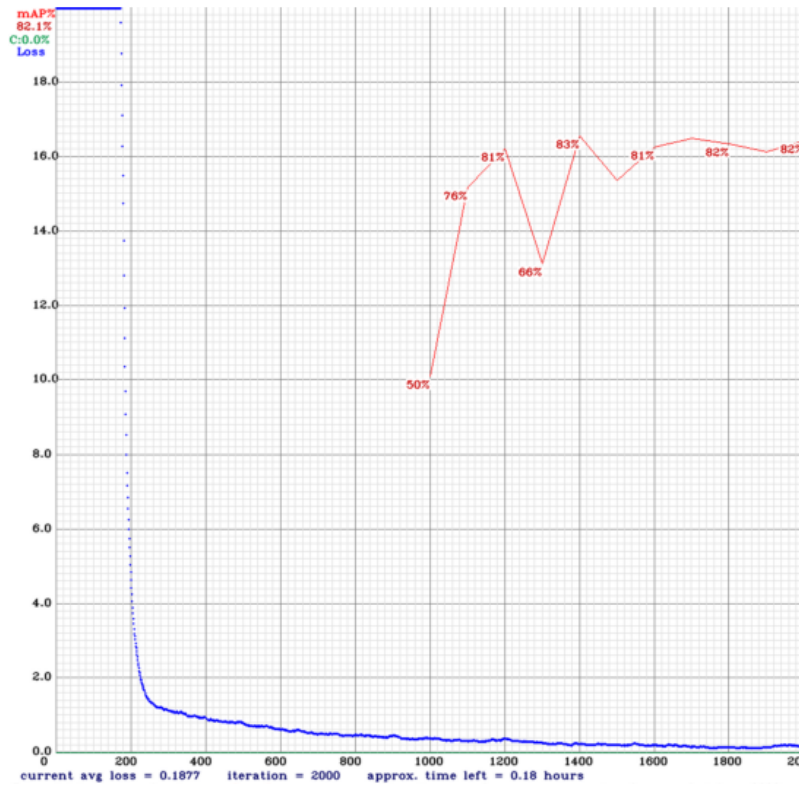


Figure 4.14 Mean average precision and IoU loss of YOLOv3

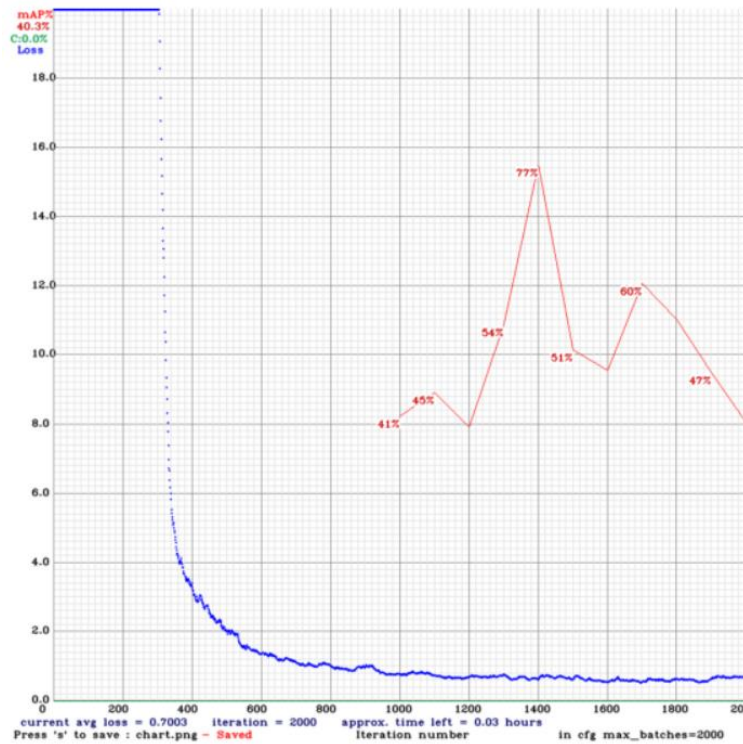


Figure 4.15 Mean average precision and IoU loss of YOLOv3-tiny

4.2.3 Experiment 3: Modified YOLOv4 with Transfer Learning and Hyperparameter Tuning

We can quickly create precise models by using a technique called transfer learning. We use a model that has previously been trained in an identical and much larger dataset to address a related problem, as opposed to initial the learning process totally from scratch. By the scope of our challenge and the quantity of our dataset, we integrate a pre-trained model, which we then fine-tune to carry out the specified task.

We put a greater emphasis on training with transfer learning using configuration and hyperparameter settings in our modified YOLOv4 version. Since there were the small number of training images, we continued to use batch sizes of 64 and subdivisions of 16. We need to alter the number of classes because we're only interested in identifying a single object, as opposed to 80 in the original configuration file, filter value and classes are changed into 18 and 1 respectively. The model is trained across 2000 batches (because at least $2000 \times \text{num classes}$ are required), with model checkpoints saved at batches 1000 and 2000 respectively.

4.2.3.1 Different learning rates

A hyperparameter that may be altered and utilized to train neural networks is the learning rate. Normally, it has a small, positive value that ranges from 0.0 to 1.0. How quickly the model adapts to a problem depends on its learning rate. Given the smaller changes to the weights made with each update, smaller learning rates necessitate more training epochs, In contrast, higher learning rates result in quicker changes and require fewer training iterations.

Table 4.4 Performance evaluation of modified YOLOv4 with different learning rates

Learning rate	Avg IoU	Precision	Recall	F1-score	mAP
0.0013	57.62	0.80	0.87	0.84	85.97%
0.0015	60.64	0.83	0.90	0.86	88.47%
0.0017	55.61	0.76	0.85	0.80	85.04%
0.0020	53.98	0.74	0.85	0.79	83.74%

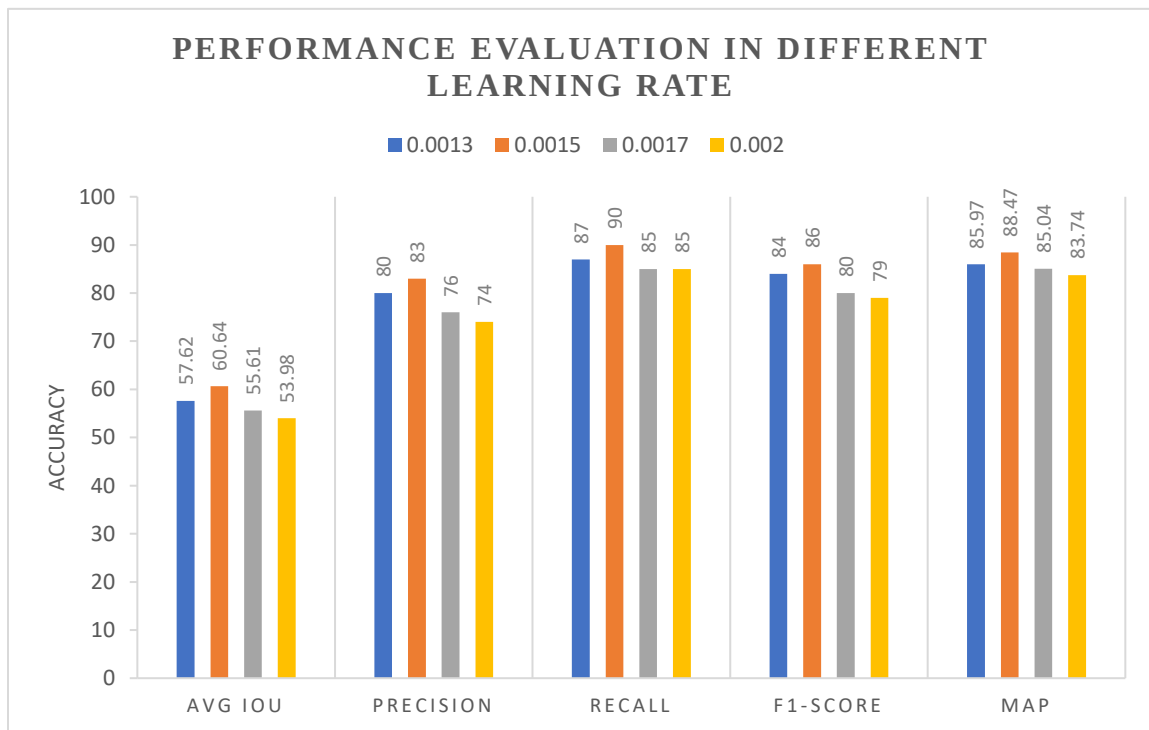


Figure 4.16 Performance evaluation for modified YOLOv4 at different learning rate

Table 4.4 and Figure 4.16 shows that, the learning rate show a high impact on its performance. To set the base value setup for the learning rate for pothole detection, several experiments were done and the 0.0015 learning rate gives the better value in every aspect with a high mean average precision value of 88.47%.



Figure 4.17 Mean average precision and IoU loss for 0.0013 learning rate

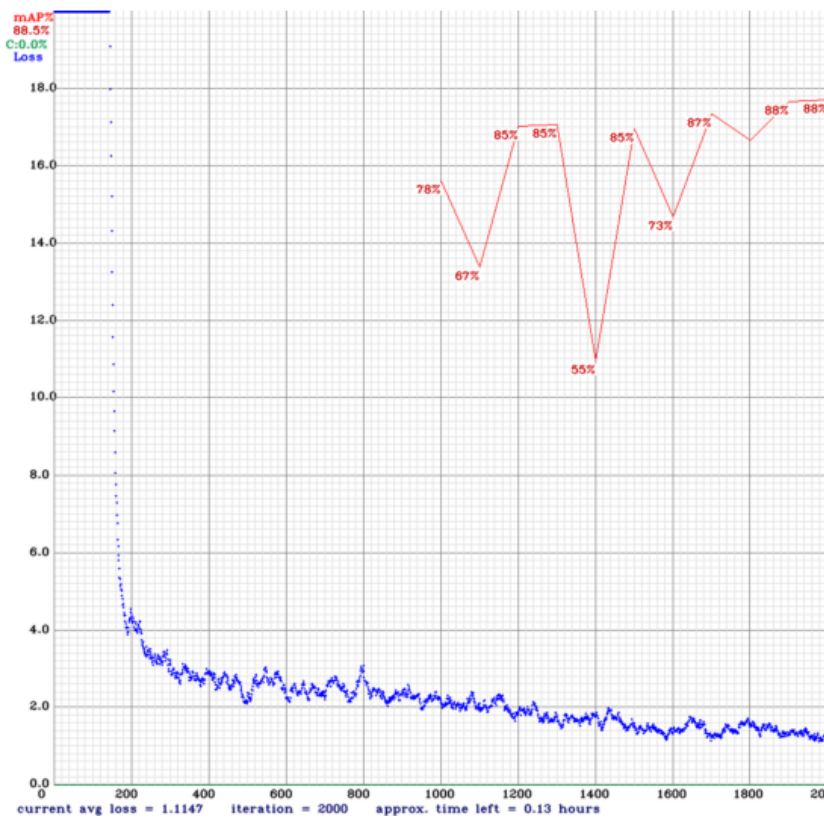


Figure 4.18 Mean average precision and IoU loss for 0.0015 learning rate

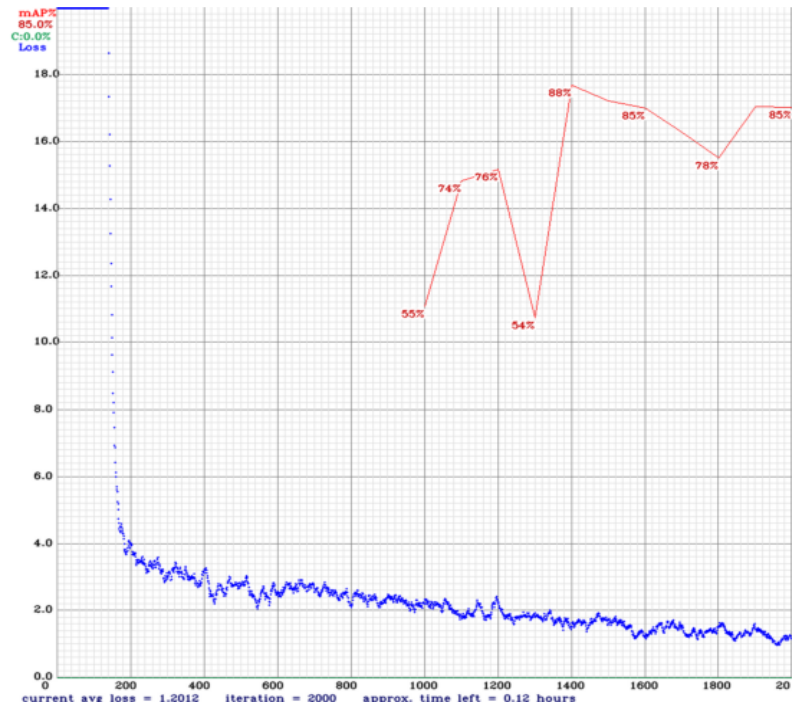


Figure 4.19 mean average precision and IoU loss for 0.0017 learning rate

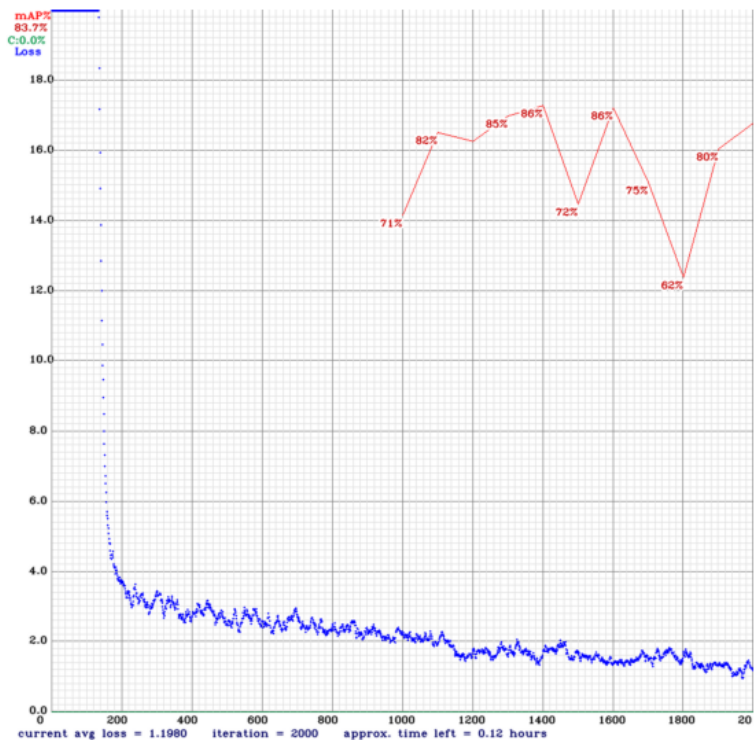


Figure 4.20 Mean average precision and IoU loss in for in 0.0020 learning rate

4.2.3.2 Hyperparameters Optimization/Adjustment

The challenge of choosing the appropriate collection of hyperparameters for a learning algorithm is referred to as hyperparameter optimization. The parameter values are initially random when a model is being trained, and they are changed as the training progresses.

For small datasets, batch and sub-division are set to 64 and 16. All different image sizes are reduced to 416×416 with channel 3 for faster execution and better result. In three layers above the YOLOv3 dense prediction layer, the filter size 18 has been set. The decay and momentum rates are 0.0005 and 0.949 respectively. The learning rate starts from 0.0015. For image saturation and exposure, both are set to 1.5. hue value is updated to 0.035. With 2000 max batches, steps also changed into 60% to 80% of its max batches with 1800,2200. In YOLOv4, IoU loss is used the Complete-IoU loss method for bounding box overlap error. Some activation like leaky, mish, and linear are used and for data augmentation, mosaic method is used in this experiment.

Table 4.5 Performance evaluation of modified YOLOv4 using a transfer learning approach

Models	Avg IoU	Precision	Recall	F1-score	mAP
Modified Yolo-v4	56.26	0.79	0.90	0.84	89.22%

After transfer learning and adjustments to various hyperparameters, modified YOLOv4 improved mean average precision over the original YOLOv4 model with 89.22% and recall of 0.90. The average IoU, precision, and f-score all generate pretty good results.

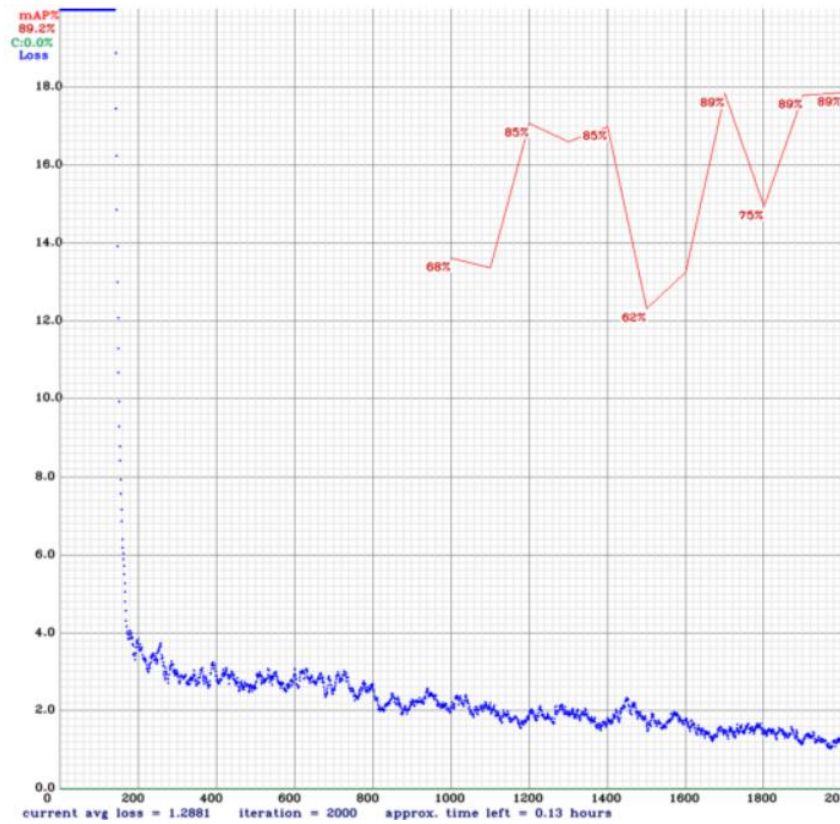


Figure 4.21 Mean average precision and IoU loss for modified YOLOv4

After calculating mean average precision and another evaluation, we tested a model with new pothole images to find the accuracy result which is shown below.







Figure 4.22 Test 2 images for testing model

In figure 4.22, Potholes in test 2 image are successfully detected with their detected percentage by creating its bounding box using modified YOLOv4. Different pothole has different detected percentage shown in the images with the bounding box according to its nature. Also, we can accurately find many potholes in a single image.

4.2.3 Experiment 4: Comparative Analysis of Similar Pothole Detection Research

With their various datasets, numerous relevant studies in the field of pothole detection have been carried out. Various researcher used various YOLO models to detect potholes in images and also tried to improve its model performance.

Table 4.5 Comparative analysis of similar pothole research

Paper	Models	Avg IoU	Precision	Recall	F1-score	mAP
Proposed Model	YOLOv4	56.26%	0.79	0.90	0.84	89.22%
(Park et al., 2020)	YOLOv4, YOLOv5	NA	NA	NA	NA	77.7%, 74.8%
(Mohd Omar and Pradeep Kumar 2020)	YOLOv4	38.38%	0.58	0.37	NA	NA
(Keval Doshi and Yasin Yilmaz 2020)	YOLOv4	NA	NA	NA	0.63	NA
(Adak et al., 2021)	YOLOv4	NA	0.65	NA	NA	NA
(Al-Shaghour et al., 2021)	YOLOv4	NA	0.85	0.81	NA	85.39%

Five recent researches were selected to compare performance in the pothole detection with our modified YOLOv4. In every aspect, our modified YOLOv4 showed the better result than the other recent studies.

CHAPTER 5: CONCLUSION AND FUTURE WORKS

Detecting potholes differs from detecting other objects like people, cars, airplanes, and so forth. Potholes don't have a definite shape like other objects do. It is difficult to raise the mean average precision for pothole detection because of the abovementioned limitation. In this research, we have collected pothole images from different region of Nepal and done a pre-processing part like data augmentation and resizing, and creating a bounding box for images. We have run the whole experiment on google colab with free GPU and we used jupyter notebook for coding.

Initially, different train/test ratios were divided to check which one is good for pothole detection. 87/13 ratio gave the better result with mean average precision of 83.44%. After setting the base value for the train/test ratio, four different YOLO variants YOLOv3, YOLOv3-tiny, YOLOv4, and YOLOv4-tiny are trained and tested with preprocessed data. YOLOv4 did a better job with 83.44%.

New features from YOLOv4, Bag-of-Specials and Bag-of-Freebies, were used for CIOU loss, Mosaic augmentation, Mish activation, etc for model training which helps to reduce IoU loss in models and also supports images augmentation for training models.

This thesis paper proposed the modified YOLOv4 using transfer learning with hyperparameter tuning which helps to modify the state-of-arts of YOLOv4 to increase its performance. Experimental results show that, when compared to other studies on pothole detection, the proposed model developed for this study has the highest detection accuracy, with a mAP of 89.22%.

Our thesis study can be extended and enhanced to make more out of it, for our work can be extended towards getting better results from our pothole detection system. In the future, we can collect more primary data and run with this model to find out its performance.

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APPENDIX

<https://www.kaggle.com/datasets/bikalbaral30/primary-pothole-images/code>

Appendix 1: Link of Pothole Images